Benchmarking Spatial Reasoning of LLMs in Text-to-3D Modeling

Bachelor's Thesis Presentation

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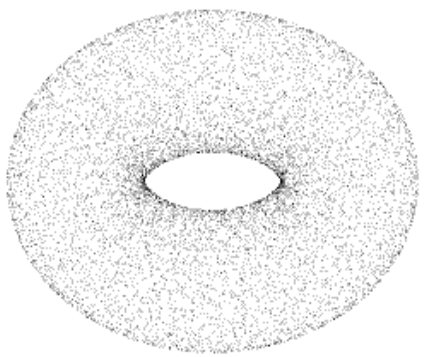
Examiner: Prof. Dr.-Ing. Klaus Diepold

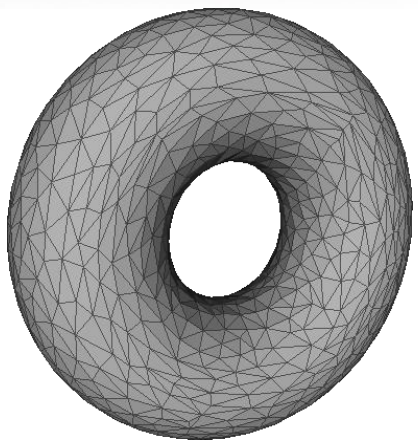
Advisor: David Fresacher, M.Sc.



Motivation

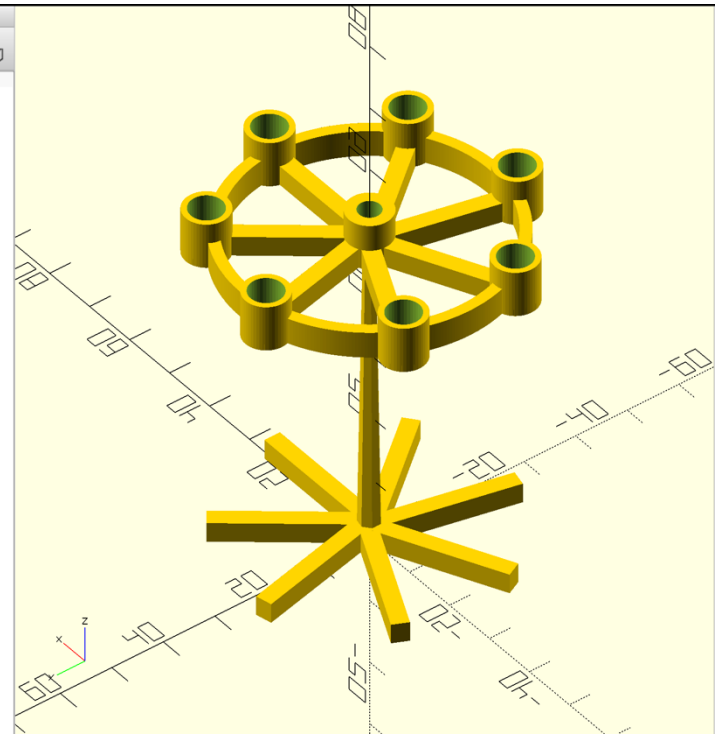






OpenScad

```
Editor
STL
1  /*[ Candle Stand ]*/
2  //Length of candle stand
3  length=50; // [70:large,50:medium,30:small]
4
5  // Center stand
6  cylinder(length,width-2);
7
8  //Radius of ring of stand
9  radius=25;
10
11 /* [ Number of candle holders ]*/
12 // Number of candle holders
13 count=7; //[3:14]
14
15 //Do you want center Candle
16 centerCandle=true;
17
18 /* [ Candle Holder ]*/
19 //Length of candle holder
20 candleSize=7;
21
22 //Width of candle holder
23 width=4;
24
25 //Size of hole for candle holder
26 holeSize=3;
27
28 CenterCandleWidth=4;
29
30 /*[Properties of support]*/
31
32 heightOfSupport=3;
33 widthOfSupport=3;
34
35 /*[Properties of Ring]*/
36
37 heightOfRing=4;
38
39 widthOfRing=23;
40
41
42 //Create center candle
43 translate([0,0,length-candleSize/2])
44 if(centerCandle){
45     difference(){
```



Research Questions

RQ1 How can we evaluate LLM-generated 3D code, and which metrics best capture its geometric accuracy?

RQ2 How do SOTA models perform in text-to-CAD generation using Zero-Shot vs. Chain-of-Thought prompting?

RQ3 How does 3D object complexity affect the success rate of procedural script generation?

Benchmark Models & Dataset

The Dataset (900 Clean Samples)

- Data Enchancement
- Validity Check
- Balanced Distribution

300	300	300
Simple	Medium	Complex

- Qwen2.5-Coder:32B
- Codestral:22B
- DeepSeek-Coder-V2-Lite

➢ Zero-Shot
➢ Chain-of-Thought (CoT)

Metrics

Intersection over Union (IoU)

$$\mathrm{IoU}(\mathrm{V}_1, V_2) = \frac{|V_1 \cap V_2|}{|V_1 \cup V_2|}$$

Chamfer Distance (CD)

$$\mathrm{d}_{CD}(S_1, S_2) = \frac{1}{|S_1|} \sum_{x \in S_1} \min_{y \in S_2} \|x - y\|_2^2 + \frac{1}{|S_2|} \sum_{y \in S_2} \min_{x \in S_1} \|x - y\|_2^2$$

Hausdorff Distance (HD)

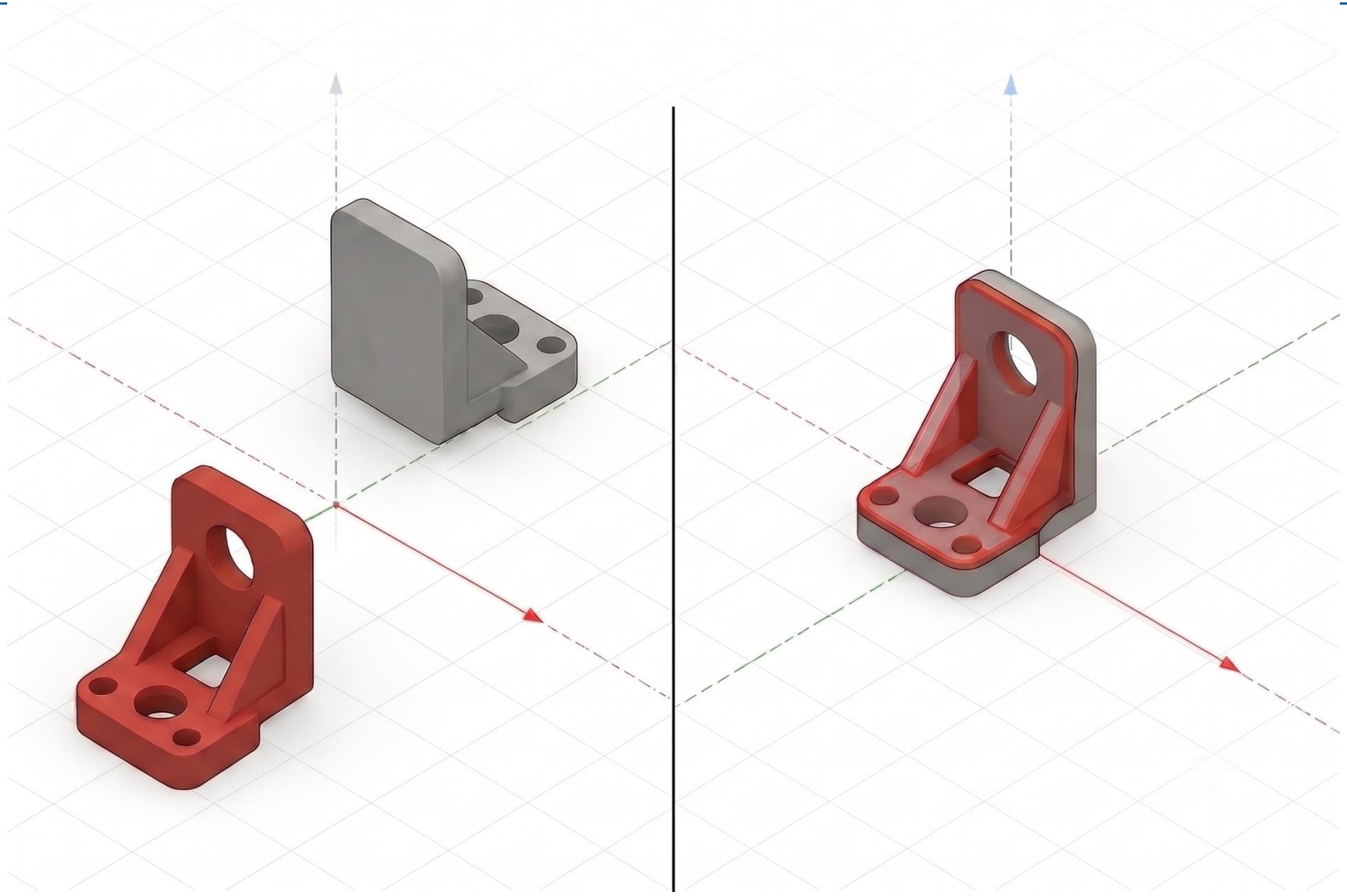
$$\mathrm{d}_{HD}(S_1, S_2) = \max\left\{\max_{x \in S_1} \min_{y \in S_2} \|x - y\|_2, \max_{y \in S_2} \min_{x \in S_1} \|x - y\|_2\right\}$$

The Composite Success Score

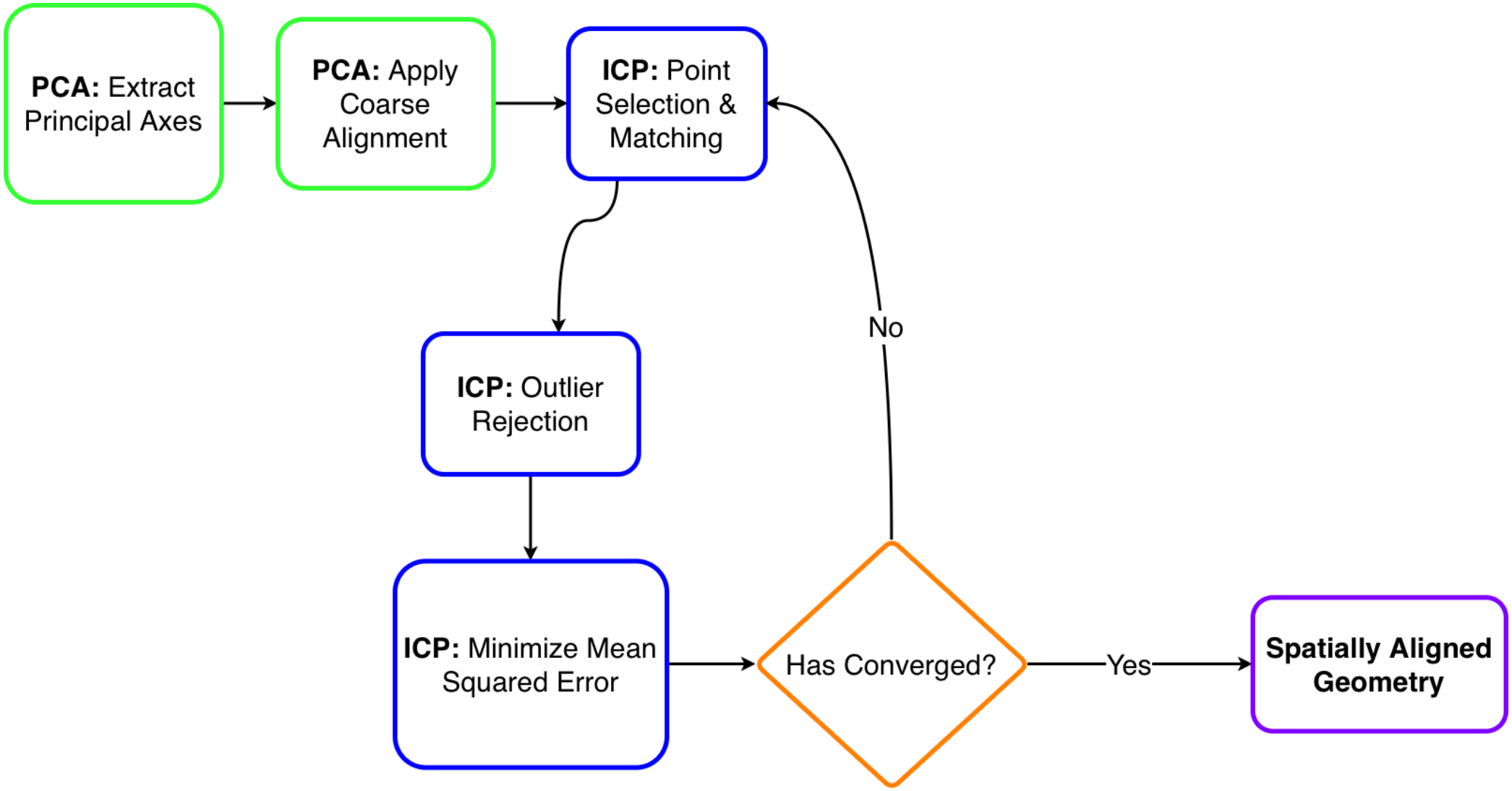
$$S_{IoU} = IoU \times 100 \qquad S_{CD} = e^{-\frac{d_{CD}}{10.0}} \times 100 \qquad S_{HD} = e^{-\frac{d_{HD}}{10.0}} \times 100$$

$$S_c = V \cdot [(0.50 \cdot S_{IoU}) + (0.25 \cdot S_{CD}) + (0.25 \cdot S_{HD})]$$

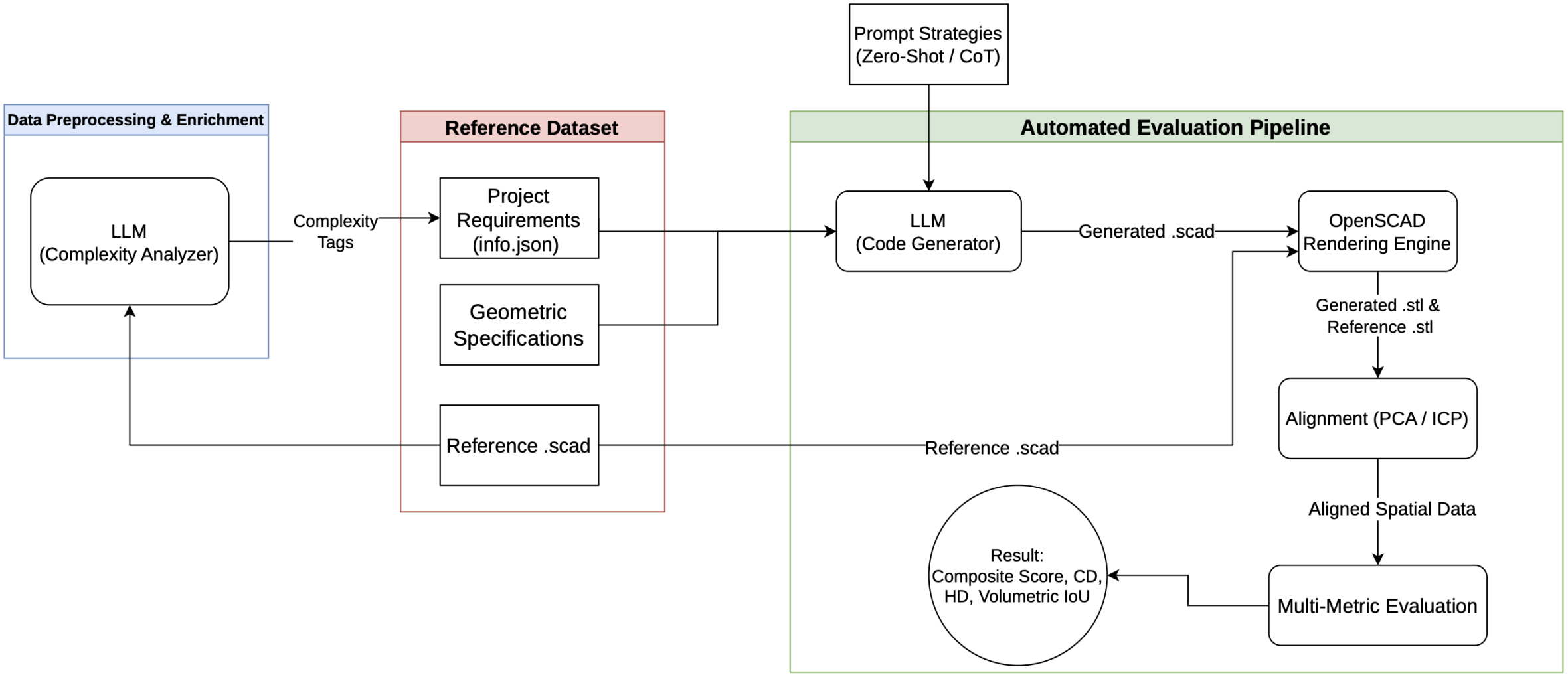
The "Alignment Problem" in Spatial Evaluation



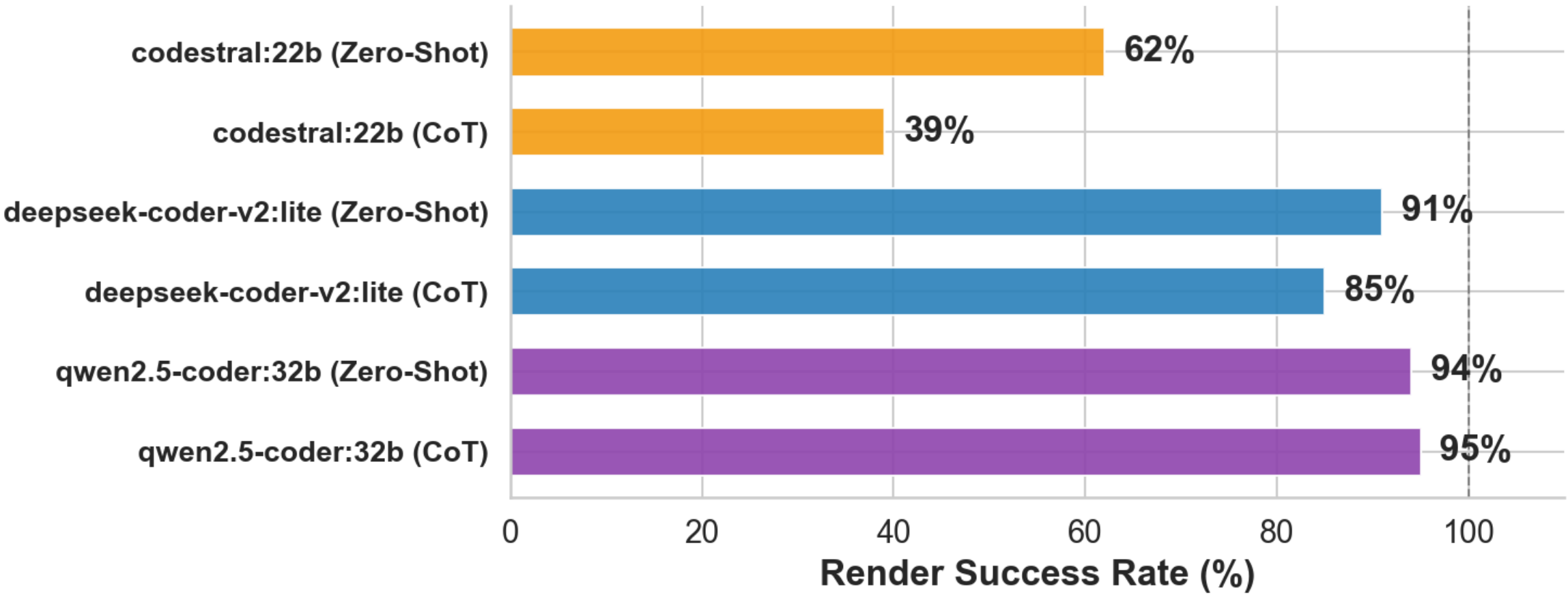
Two-stage Spatial Alignment



The Pipeline

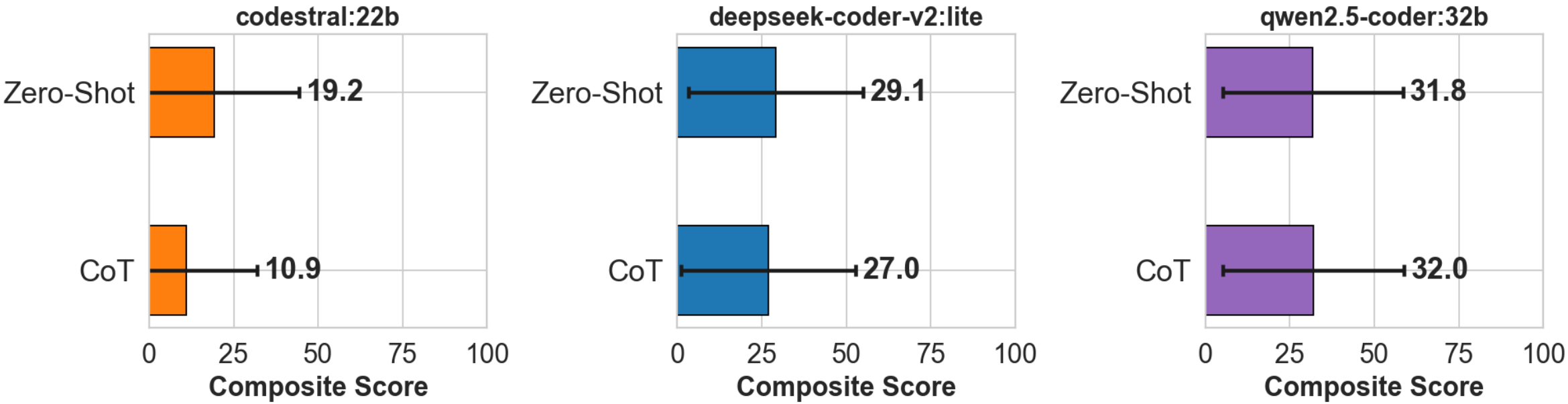


Result: Rendering Validity



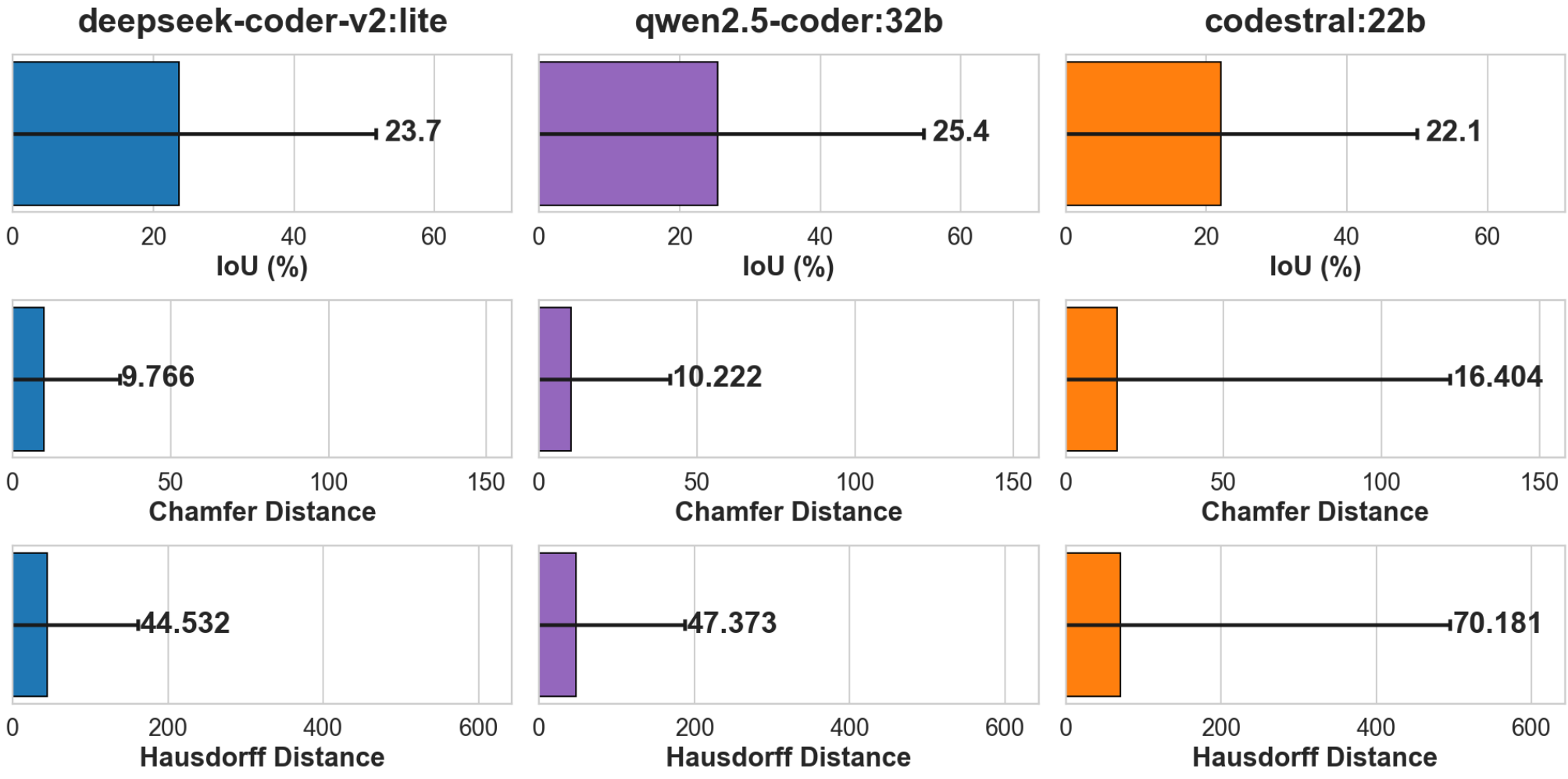
Key Takeaway: Qwen leads overall, while CoT surprisingly degrades performance in smaller models.

Result: Composite Success Score



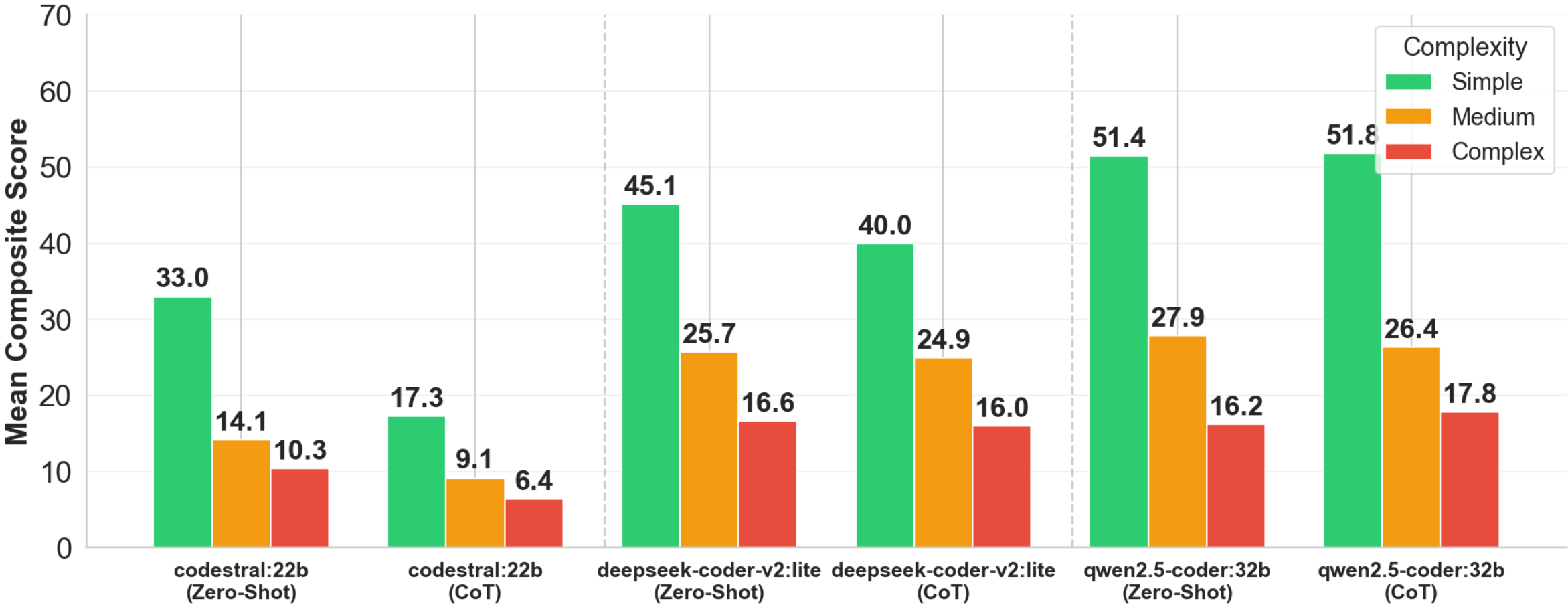
Key Takeaway: CoT acts as a destabilizing factor for smaller models, dropping Codestral's validity 39% due to syntax overhead.

Result: Metrics (Zero-Shot)



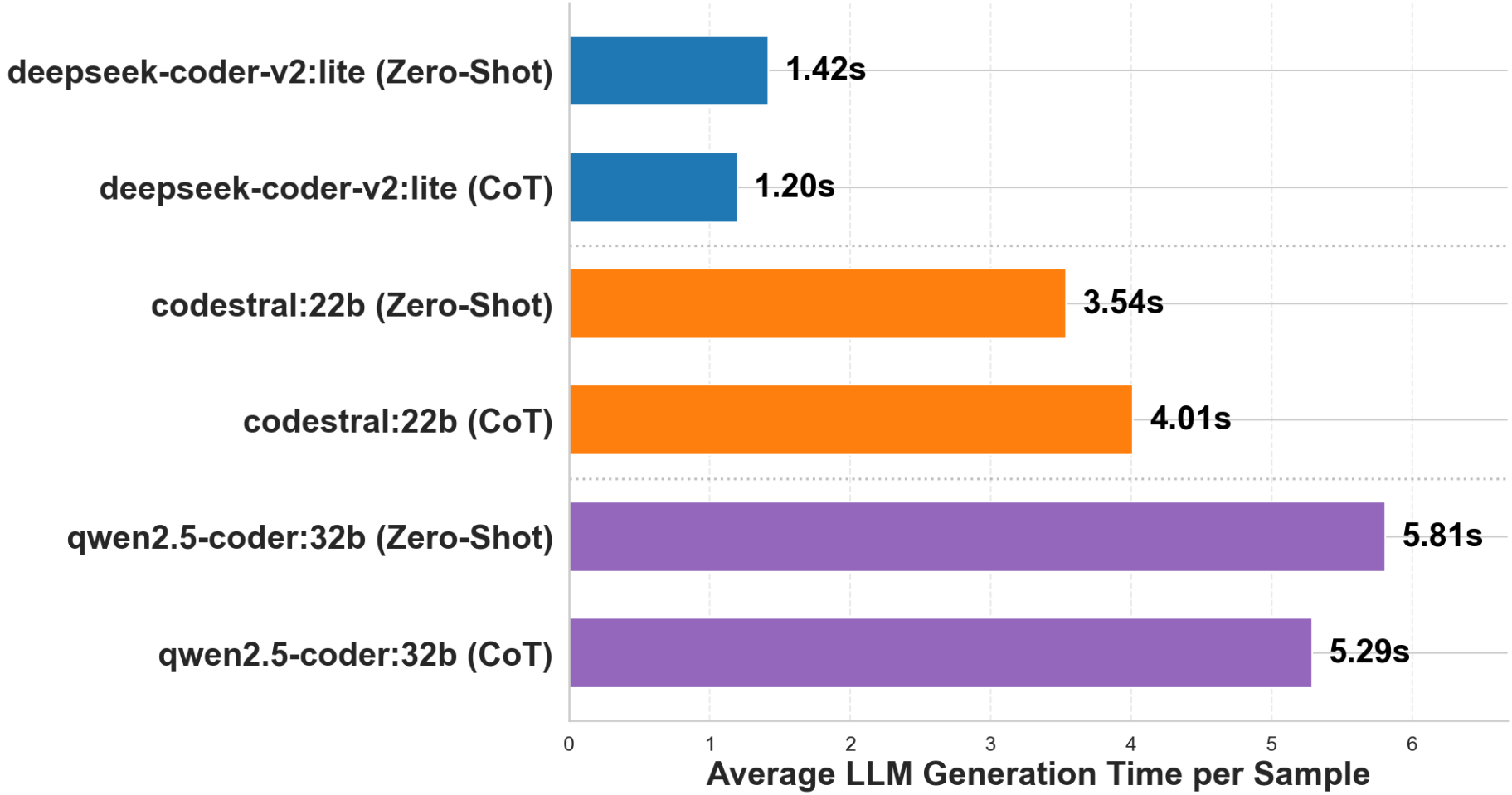
Key Takeaway: Qwen leads in volume overlap, while DeepSeek slightly edges out in surface accuracy.

Result: Impact of Complexity



Key Takeaway: Geometric complexity severely degrades performance across all models.

Result: LLM Generation Time

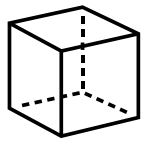


Key Takeaway: DeepSeek's lightweight MoE architecture offers the fastest generation, while Qwen trades inference speed for maximum structural validity.

Conclusion



Without **PCA + ICP** registration, standard metrics unfairly penalize coordinate misplacements. A **Multi-Metric** approach is required to capture unique AI failure modes.



While LLMs can generate valid OpenSCAD syntax perfectly, their **"Euclidean" spatial logic** fails rapidly on complex shapes with **nested dependencies**.



SOTA models excel at syntax, yet volumetric and surface metrics show they still struggle with **exact shape fidelity**. There is room for improvement.

Thank you for listening.